



Question Bank

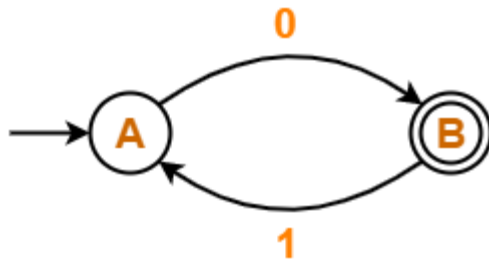
Theory Of Automata & Formal Languages (Dr. A.P.J. Abdul Kalam Technical University)



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Problem-01:

Find regular expression for the following DFA using Arden's Theorem-



Solution-

Step-01:

Form an equation for each state-

$$A = \epsilon + B.1 \dots\dots(1)$$

$$B = A.0 \dots\dots(2)$$

Step-02:

Bring final state in the form $R = Q + RP$.

Using (1) in (2), we get-

$$B = (\epsilon + B.1).0$$

$$B = \epsilon.0 + B.1.0$$

$$B = 0 + B.(1.0) \dots\dots(3)$$

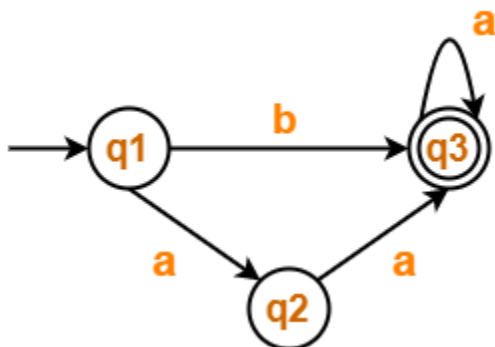
Using Arden's Theorem in (3), we get-

$$B = 0.(1.0)^*$$

Thus, Regular Expression for the given DFA = $0(10)^*$

Problem-02:

Find regular expression for the following DFA using Arden's Theorem-



Solution-

Step-01:

Form an equation for each state-

$$q1 = \epsilon \dots\dots(1)$$

$$q2 = q1.a \dots\dots(2)$$

$$q3 = q1.b + q2.a + q3.a \dots\dots(3)$$

Step-02:

Bring final state in the form $R = Q + RP$.

Using (1) in (2), we get-

$$q_2 = \epsilon.a$$

$$q_2 = a \dots\dots(4)$$

Using (1) and (4) in (3), we get-

$$q_3 = q_1.b + q_2.a + q_3.a$$

$$q_3 = \epsilon.b + a.a + q_3.a$$

$$q_3 = (b + a.a) + q_3.a \dots\dots(5)$$

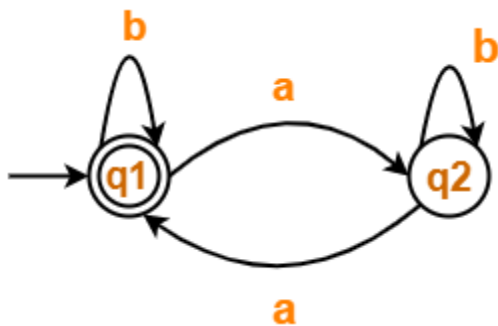
Using Arden's Theorem in (5), we get-

$$q_3 = (b + a.a)a^*$$

Thus, Regular Expression for the given DFA = $(b + aa)a^*$

Problem-03:

Find regular expression for the following DFA using Arden's Theorem-



Solution-

Step-01:

Form an equation for each state-

$$q_1 = \epsilon + q_1.b + q_2.a \dots\dots(1)$$

$$q_2 = q_1.a + q_2.b \dots\dots(2)$$

Step-02:

Bring final state in the form $R = Q + RP$.

Using Arden's Theorem in (2), we get-

$$q_2 = q_1.a.b^* \dots\dots(3)$$

Using (3) in (1), we get-

$$q_1 = \epsilon + q_1.b + q_1.a.b^*.a$$

$$q_1 = \epsilon + q_1.(b + a.b^*.a) \dots\dots(4)$$

Using Arden's Theorem in (4), we get-

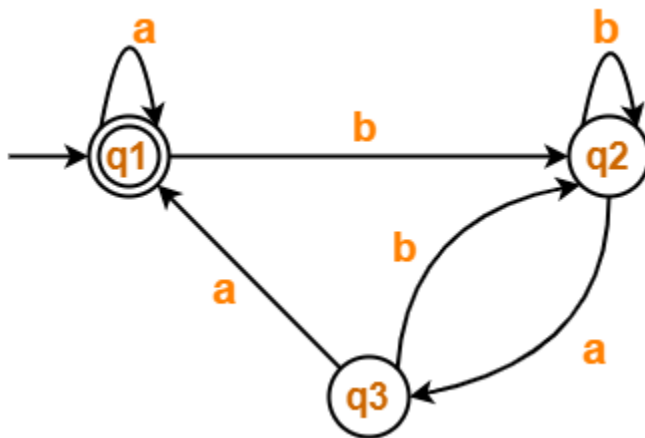
$$q1 = \epsilon.(b + a.b^*.a)^*$$

$$q1 = (b + a.b^*.a)^*$$

Thus, Regular Expression for the given DFA = $(b + a.b^*.a)^*$

Problem-04:

Find regular expression for the following DFA using Arden's Theorem-



Solution-

Step-01:

Form an equation for each state-

$$q1 = \epsilon + q1.a + q3.a \dots\dots(1)$$

$$q2 = q1.b + q2.b + q3.b \dots\dots(2)$$

$$q3 = q2.a \dots\dots(3)$$

Step-02:

Bring final state in the form $R = Q + RP$.

Using (3) in (2), we get-

$$q2 = q1.b + q2.b + q2.a.b$$

$$q2 = q1.b + q2.(b + a.b) \dots\dots(4)$$

Using Arden's Theorem in (4), we get-

$$q2 = q1.b.(b + a.b)^* \dots\dots(5)$$

Using (5) in (3), we get-

$$q_3 = q_1.b.(b + a.b)^*.a \dots\dots(6)$$

Using (6) in (1), we get-

$$q_1 = \epsilon + q_1.a + q_1.b.(b + a.b)^*.a.a$$

$$q_1 = \epsilon + q_1.(a + b.(b + a.b)^*.a.a) \dots\dots(7)$$

Using Arden's Theorem in (7), we get-

$$q_1 = \epsilon.(a + b.(b + a.b)^*.a.a)^*$$

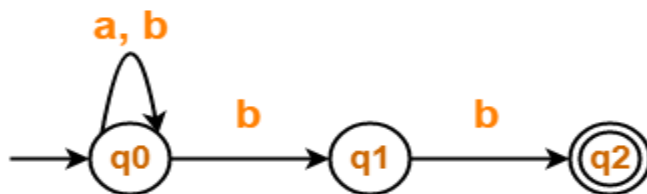
$$q_1 = (a + b.(b + a.b)^*.a.a)^*$$

Thus, Regular Expression for the given DFA = $(a + b(b + ab)^*aa)^*$

NFA TO DFA CONVERSION

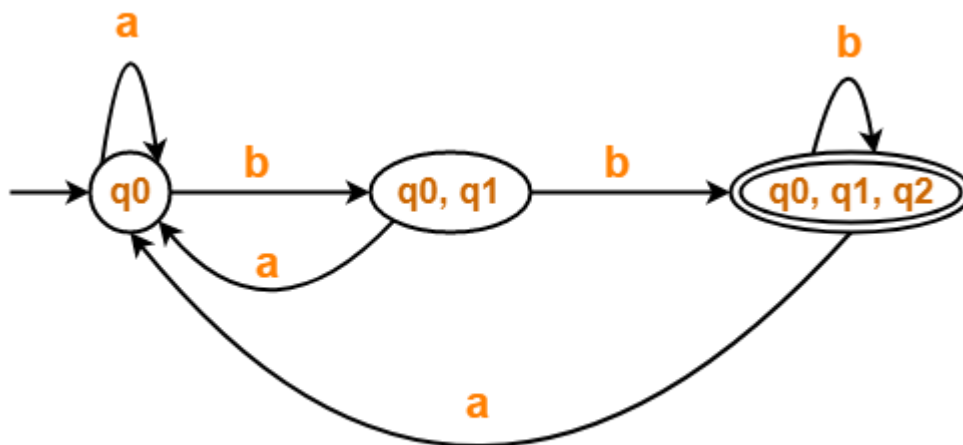
Problem-01:

Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA)-



Solution-

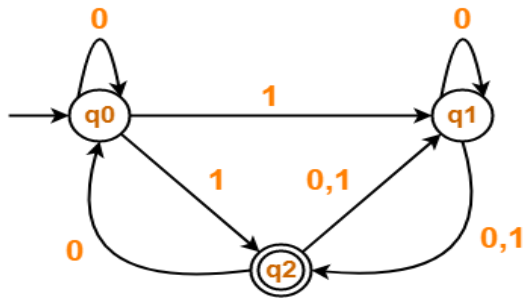
Deterministic Finite Automata (DFA) may be drawn as-



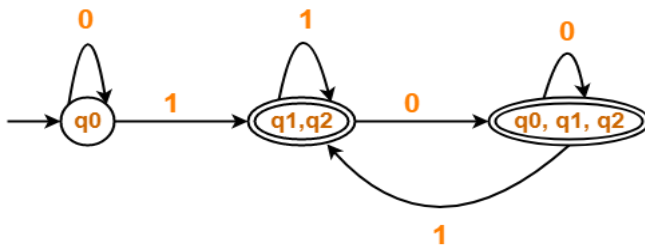
Deterministic Finite Automata (DFA)

Problem-02:

Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA)-



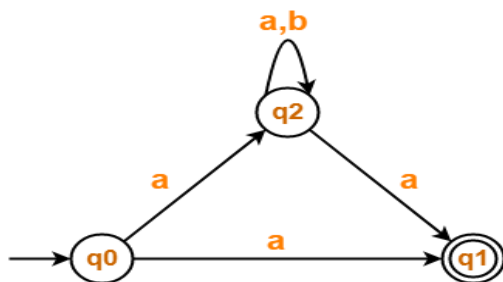
Solution-Deterministic Finite Automata (DFA) may be drawn as-



Deterministic Finite Automata (DFA)

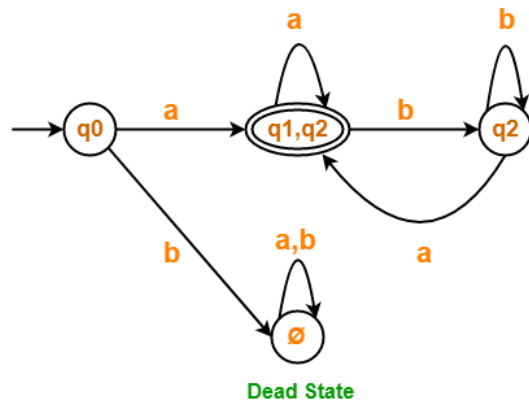
Problem-03:

Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA)-



Solution-

Now, Deterministic Finite Automata (DFA) may be drawn as-

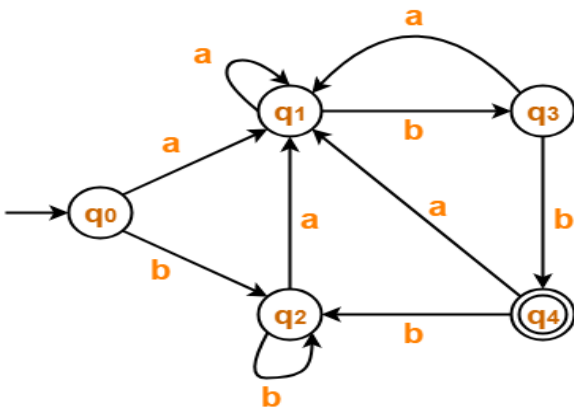


Deterministic Finite Automata (DFA)

DFA MINIMIZATION

Problem-01:

Minimize the given DFA-



Solution-

Step-01:

The given DFA contains no dead states and inaccessible states.

Step-02:

Draw a state transition table-

	a	b
→q0	q1	q2
q1	q1	q3
q2	q1	q2
q3	q1	*q4
*q4	q1	q2

Step-03:

Now using Equivalence Theorem, we have-

$P_0 = \{q_0, q_1, q_2, q_3\} \{q_4\}$

$P_1 = \{q_0, q_1, q_2\} \{q_3\} \{q_4\}$

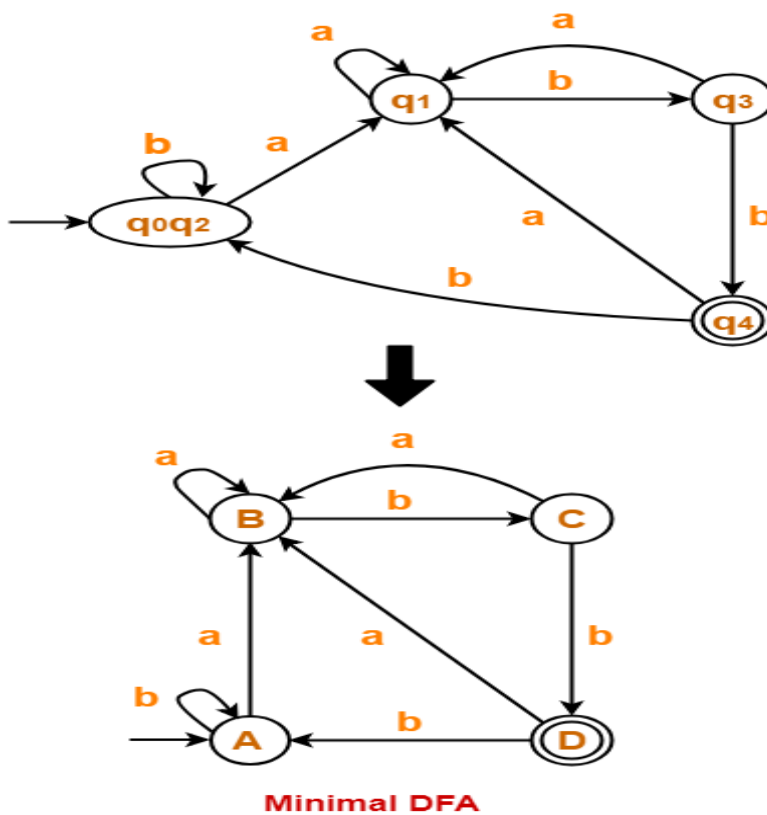
$P_2 = \{q_0, q_2\} \{q_1\} \{q_3\} \{q_4\}$

$P_3 = \{q_0, q_2\} \{q_1\} \{q_3\} \{q_4\}$

Since $P_3 = P_2$, so we stop.

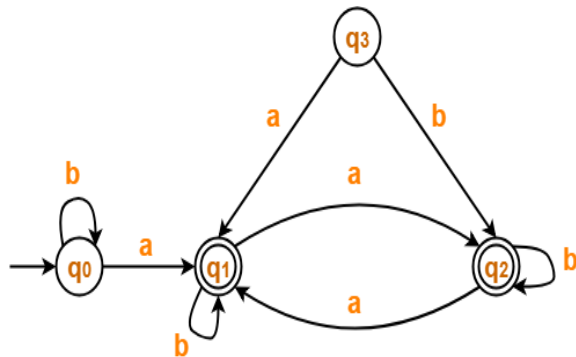
From P_3 , we infer that states q_0 and q_2 are equivalent and can be merged together.

So, Our minimal DFA is-



Problem-02:

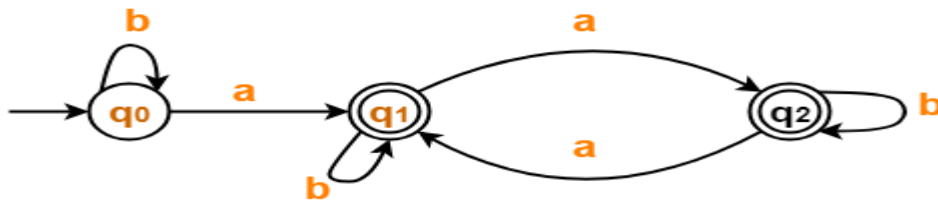
Minimize the given DFA-

**Solution-****Step-01:**

State q_3 is inaccessible from the initial state.

So, we eliminate it and its associated edges from the DFA.

The resulting DFA is-

**Step-02:**

Draw a state transition table-

	a	b
$\rightarrow q_0$	* q_1	q_0
* q_1	* q_2	* q_1
* q_2	* q_1	* q_2

Step-03:

Now using Equivalence Theorem, we have-

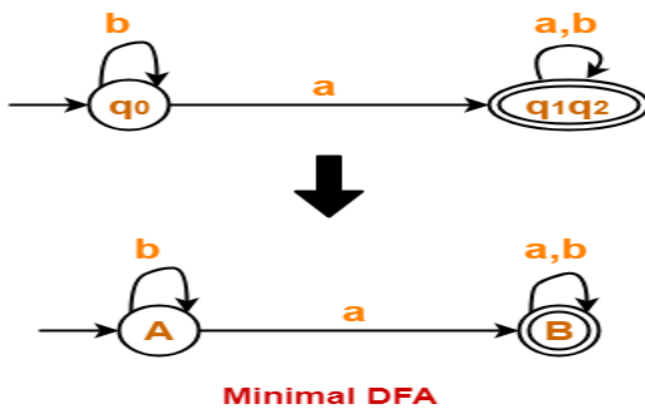
$$P_0 = \{ q_0 \} \{ q_1, q_2 \}$$

$$P_1 = \{ q_0 \} \{ q_1, q_2 \}$$

Since $P_1 = P_0$, so we stop.

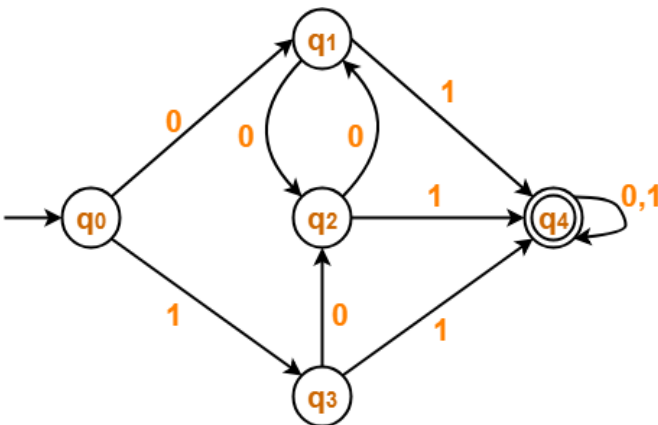
From P_1 , we infer that states q_1 and q_2 are equivalent and can be merged together.

So, Our minimal DFA is-



Problem-03:

Minimize the given DFA-



Solution-

Step-01:

The given DFA contains no dead states and inaccessible states.

Step-02:

Draw a state transition table-

	0	1
→q0	q1	q3
q1	q2	*q4
q2	q1	*q4
q3	q2	*q4
*q4	*q4	*q4

Step-03:

Now using Equivalence Theorem, we have-

$$P_0 = \{ q_0, q_1, q_2, q_3 \} \{ q_4 \}$$

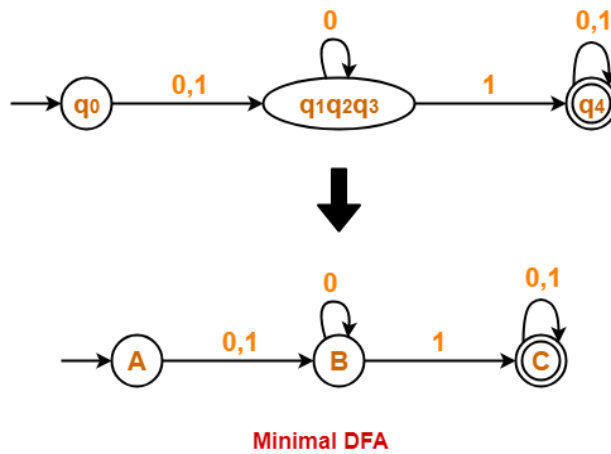
$$P_1 = \{ q_0 \} \{ q_1, q_2, q_3 \} \{ q_4 \}$$

$$P_2 = \{ q_0 \} \{ q_1, q_2, q_3 \} \{ q_4 \}$$

Since $P_2 = P_1$, so we stop.

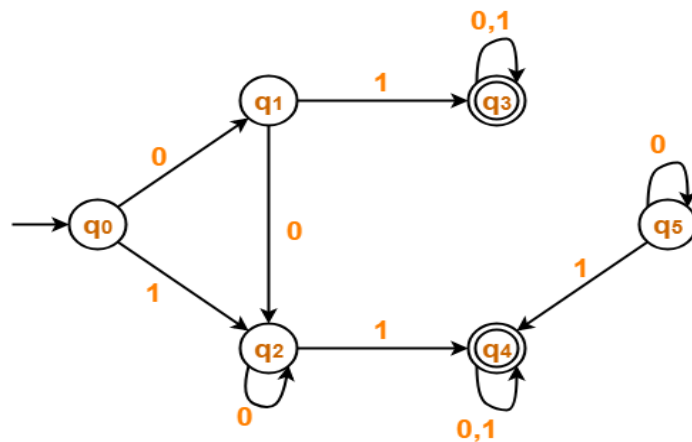
From P_2 , we infer that states q_1 , q_2 and q_3 are equivalent and can be merged together.

So, Our minimal DFA is-



Problem-04:

Minimize the given DFA-



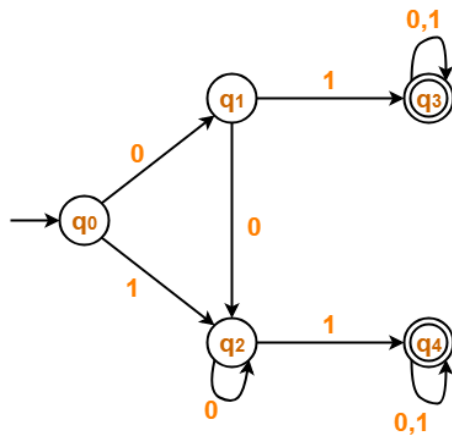
Solution-

Step-01:

State q_5 is inaccessible from the initial state.

So, we eliminate it and its associated edges from the DFA.

The resulting DFA is-



Step-02:

Draw a state transition table-

	0	1
→q0	q1	q2
q1	q2	*q3
q2	q2	*q4
*q3	*q3	*q3
*q4	*q4	*q4

Step-03:

Now using Equivalence Theorem, we have-

$$P_0 = \{ q_0, q_1, q_2 \} \{ q_3, q_4 \}$$

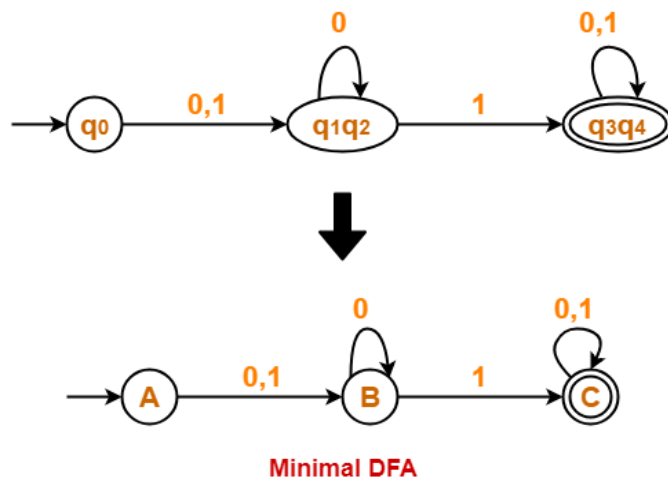
$$P_1 = \{ q_0 \} \{ q_1, q_2 \} \{ q_3, q_4 \}$$

$$P_2 = \{ q_0 \} \{ q_1, q_2 \} \{ q_3, q_4 \}$$

Since $P_2 = P_1$, so we stop.

From P_2 , we infer-

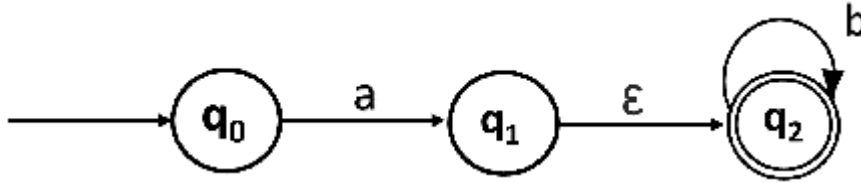
States q_1 and q_2 are equivalent and can be merged together. States q_3 and q_4 are equivalent and can be merged together. So, our minimal DFA is-



NFA with ϵ -moves to NFA without ϵ -moves

Example:

Convert the following NFA with ϵ to NFA without ϵ .



Solutions: We will first obtain ϵ -closures of q_0 , q_1 and q_2 as follows:

$$\epsilon\text{-closure}(q_0) = \{q_0\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

$$\begin{aligned}\delta'(q_0, a) &= \epsilon\text{-closure}(\delta(\delta^{\wedge}(q_0, \epsilon), a)) \\ &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, a)) \\ &= \epsilon\text{-closure}(q_1) \\ &= \{q_1, q_2\}\end{aligned}$$

$$\begin{aligned}\delta'(q_0, b) &= \epsilon\text{-closure}(\delta(\delta^{\wedge}(q_0, \epsilon), b)) \\ &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, b)) \\ &= \Phi\end{aligned}$$

Now the δ' transition on q_1 is obtained as:

$$\begin{aligned}\delta'(q_1, a) &= \epsilon\text{-closure}(\delta(\delta^{\wedge}(q_1, \epsilon), a)) \\ &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), a)) \\ &= \epsilon\text{-closure}(\delta(q_1, q_2), a) \\ &= \epsilon\text{-closure}(\delta(q_1, a) \cup \delta(q_2, a)) \\ &= \epsilon\text{-closure}(\Phi \cup \Phi) \\ &= \Phi\end{aligned}$$

$$\begin{aligned}\delta'(q_1, b) &= \epsilon\text{-closure}(\delta(\delta^{\wedge}(q_1, \epsilon), b)) \\ &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), b)) \\ &= \epsilon\text{-closure}(\delta(q_1, q_2), b) \\ &= \epsilon\text{-closure}(\delta(q_1, b) \cup \delta(q_2, b)) \\ &= \epsilon\text{-closure}(\Phi \cup q_2) \\ &= \{q_2\}\end{aligned}$$

The δ' transition on q_2 is obtained as:

$$\begin{aligned}\delta'(q_2, a) &= \epsilon\text{-closure}(\delta(\delta^{\wedge}(q_2, \epsilon), a)) \\ &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), a)) \\ &= \epsilon\text{-closure}(\delta(q_2, a)) \\ &= \epsilon\text{-closure}(\Phi) \\ &= \Phi\end{aligned}$$

$$\begin{aligned}
 \delta'(q_2, b) &= \epsilon\text{-closure}(\delta(\delta^*(q_2, \epsilon), b)) \\
 &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), b)) \\
 &= \epsilon\text{-closure}(\delta(q_2, b)) \\
 &= \epsilon\text{-closure}(q_2) \\
 &= \{q_2\}
 \end{aligned}$$

Now we will summarize all the computed δ' transitions:

$$\delta'(q_0, a) = \{q_0, q_1\}$$

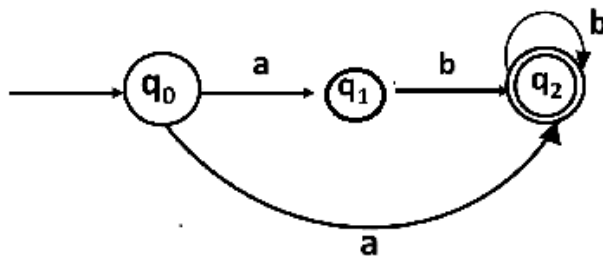
$$\delta'(q_0, b) = \Phi$$

$$\delta'(q_1, a) = \Phi$$

$$\delta'(q_1, b) = \{q_2\}$$

$$\delta'(q_2, a) = \Phi$$

$$\delta'(q_2, b) = \{q_2\}$$



NFA with ϵ -moves to DFA

Steps for converting NFA with ϵ to DFA:

Step 1: We will take the ϵ -closure for the starting state of NFA as a starting state of DFA.

Step 2: Find the states for each input symbol that can be traversed from the present. That means the union of transition value and their closures for each state of NFA present in the current state of DFA.

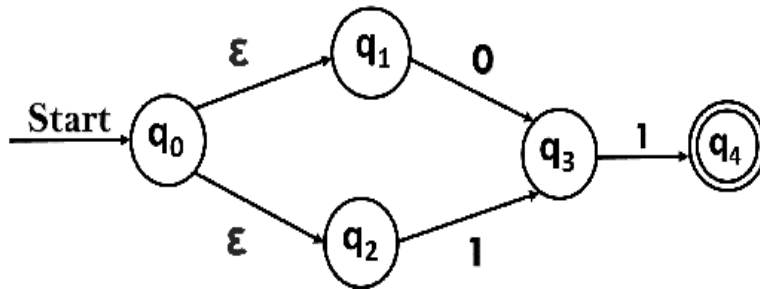
Step 3: If we found a new state, take it as current state and repeat step 2.

Step 4: Repeat Step 2 and Step 3 until there is no new state present in the transition table of DFA.

Step 5: Mark the states of DFA as a final state which contains the final state of NFA.

Example 1:

Convert the NFA with ϵ into its equivalent DFA.



Solution:

Let us obtain ϵ -closure of each state.

1. ϵ -closure $\{q_0\} = \{q_0, q_1, q_2\}$
2. ϵ -closure $\{q_1\} = \{q_1\}$
3. ϵ -closure $\{q_2\} = \{q_2\}$
4. ϵ -closure $\{q_3\} = \{q_3\}$
5. ϵ -closure $\{q_4\} = \{q_4\}$

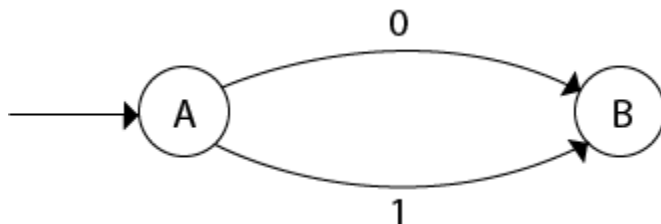
Now, let ϵ -closure $\{q_0\} = \{q_0, q_1, q_2\}$ be state A.

Hence

$$\begin{aligned}\delta'(A, 0) &= \epsilon\text{-closure} \{ \delta((q_0, q_1, q_2), 0) \} \\ &= \epsilon\text{-closure} \{ \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0) \} \\ &= \epsilon\text{-closure} \{q_3\} \\ &= \{q_3\} \quad \text{call it as state B.}\end{aligned}$$

$$\begin{aligned}\delta'(A, 1) &= \epsilon\text{-closure} \{ \delta((q_0, q_1, q_2), 1) \} \\ &= \epsilon\text{-closure} \{ \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1) \} \\ &= \epsilon\text{-closure} \{q_3\} \\ &= \{q_3\} = B.\end{aligned}$$

The partial DFA will be



Now,

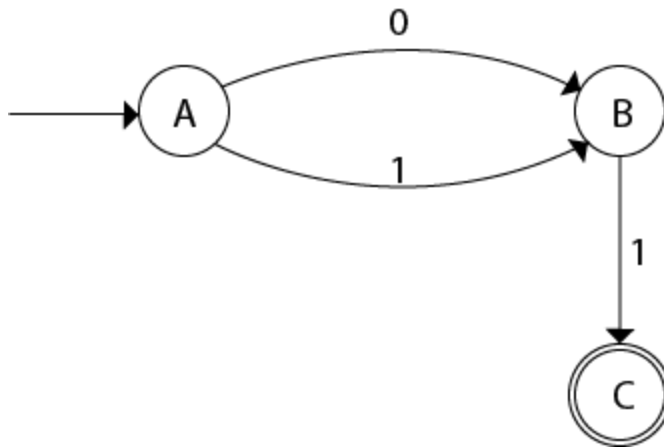
$$\delta'(B, 0) = \epsilon\text{-closure} \{ \delta(q_3, 0) \}$$

$$\begin{aligned}
 &= \phi \\
 \delta'(B, 1) &= \varepsilon\text{-closure} \{ \delta(q3, 1) \} \\
 &= \varepsilon\text{-closure} \{ q4 \} \\
 &= \{ q4 \} \quad \text{i.e. state C}
 \end{aligned}$$

For state C:

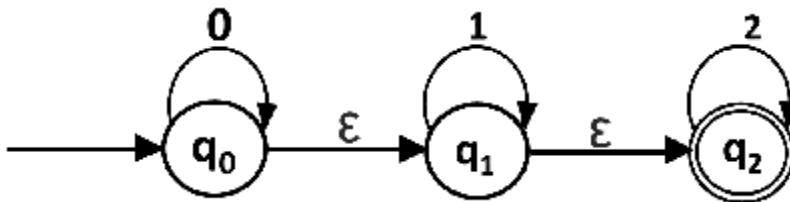
1. $\delta'(C, 0) = \varepsilon\text{-closure} \{ \delta(q4, 0) \}$
2. $= \phi$
3. $\delta'(C, 1) = \varepsilon\text{-closure} \{ \delta(q4, 1) \}$
4. $= \phi$

The DFA will be,



Example 2:

Convert the given NFA into its equivalent DFA.



Solution: Let us obtain the ε -closure of each state.

1. $\varepsilon\text{-closure}(q0) = \{q0, q1, q2\}$
2. $\varepsilon\text{-closure}(q1) = \{q1, q2\}$
3. $\varepsilon\text{-closure}(q2) = \{q2\}$

Now we will obtain δ' transition. Let $\varepsilon\text{-closure}(q0) = \{q0, q1, q2\}$ call it as **state A**.

$$\begin{aligned}
 \delta'(A, 0) &= \varepsilon\text{-closure} \{ \delta(q0, 0), \delta(q1, 0), \delta(q2, 0) \} \\
 &= \varepsilon\text{-closure} \{ q0, q1, q2 \} \\
 &= \varepsilon\text{-closure} \{ q0 \}
 \end{aligned}$$

$$= \{q_0, q_1, q_2\}$$

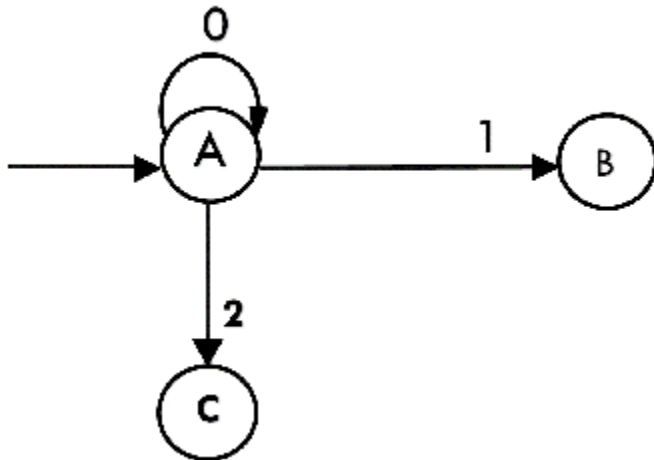
$$\begin{aligned}\delta'(A, 1) &= \varepsilon\text{-closure}\{\delta((q_0, q_1, q_2), 1)\} \\ &= \varepsilon\text{-closure}\{\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)\} \\ &= \varepsilon\text{-closure}\{q_1\} \\ &= \{q_1, q_2\} \quad \text{call it as state B}\end{aligned}$$

$$\begin{aligned}\delta'(A, 2) &= \varepsilon\text{-closure}\{\delta((q_0, q_1, q_2), 2)\} \\ &= \varepsilon\text{-closure}\{\delta(q_0, 2) \cup \delta(q_1, 2) \cup \delta(q_2, 2)\} \\ &= \varepsilon\text{-closure}\{q_2\} \\ &= \{q_2\} \quad \text{call it state C}\end{aligned}$$

Thus we have obtained

1. $\delta'(A, 0) = A$
2. $\delta'(A, 1) = B$
3. $\delta'(A, 2) = C$

The partial DFA will be:



Now we will find the transitions on states B and C for each input.

Hence

$$\begin{aligned}\delta'(B, 0) &= \varepsilon\text{-closure}\{\delta((q_1, q_2), 0)\} \\ &= \varepsilon\text{-closure}\{\delta(q_1, 0) \cup \delta(q_2, 0)\} \\ &= \varepsilon\text{-closure}\{\phi\} \\ &= \phi\end{aligned}$$

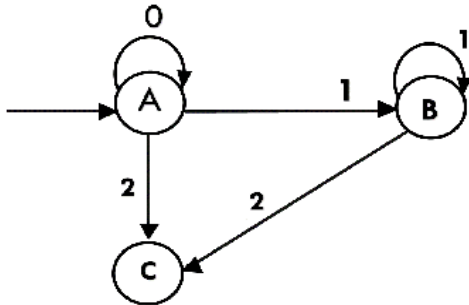
$$\begin{aligned}\delta'(B, 1) &= \varepsilon\text{-closure}\{\delta((q_1, q_2), 1)\} \\ &= \varepsilon\text{-closure}\{\delta(q_1, 1) \cup \delta(q_2, 1)\} \\ &= \varepsilon\text{-closure}\{q_1\} \\ &= \{q_1, q_2\} \quad \text{i.e. state B itself}\end{aligned}$$

$$\begin{aligned}\delta'(B, 2) &= \varepsilon\text{-closure}\{\delta((q_1, q_2), 2)\} \\ &= \varepsilon\text{-closure}\{\delta(q_1, 2) \cup \delta(q_2, 2)\} \\ &= \varepsilon\text{-closure}\{q_2\} \\ &= \{q_2\} \quad \text{i.e. state C itself}\end{aligned}$$

Thus we have obtained

1. $\delta'(B, 0) = \phi$
2. $\delta'(B, 1) = B$
3. $\delta'(B, 2) = C$

The partial transition diagram will be



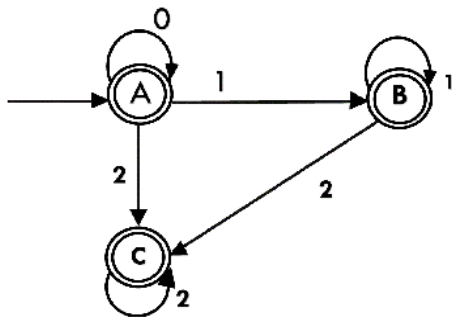
Now we will obtain transitions for C:

$$\begin{aligned}\delta'(C, 0) &= \varepsilon\text{-closure}\{\delta(q_2, 0)\} \\ &= \varepsilon\text{-closure}\{\phi\} \\ &= \phi\end{aligned}$$

$$\begin{aligned}\delta'(C, 1) &= \varepsilon\text{-closure}\{\delta(q_2, 1)\} \\ &= \varepsilon\text{-closure}\{\phi\} \\ &= \phi\end{aligned}$$

$$\begin{aligned}\delta'(C, 2) &= \varepsilon\text{-closure}\{\delta(q_2, 2)\} \\ &= \{q_2\}\end{aligned}$$

Hence the DFA is

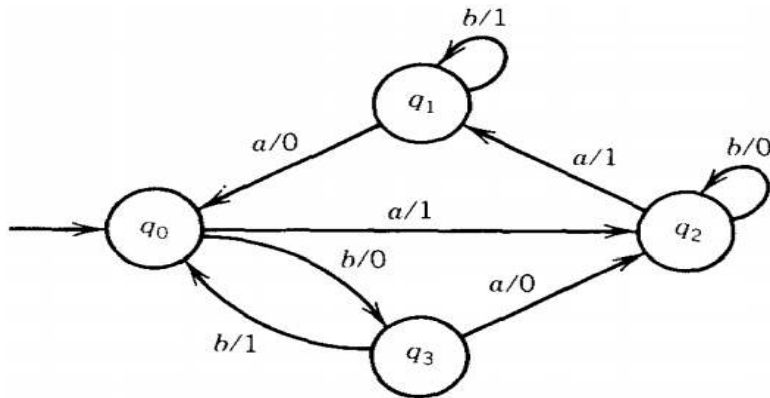


As $A = \{q_0, q_1, q_2\}$ in which final state q_2 lies hence A is final state. $B = \{q_1, q_2\}$ in which the state q_2 lies hence B is also final state. $C = \{q_2\}$, the state q_2 lies hence C is also a final state.

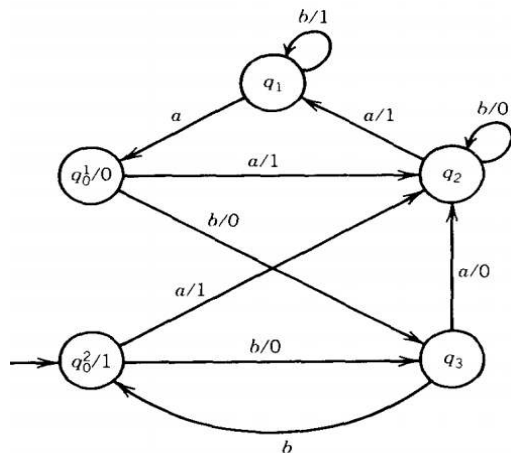
MEALY MACHINE TO MOORE MACHINE CONVERSION

Example 1:

Convert the following Mealy machine into equivalent Moore machine.

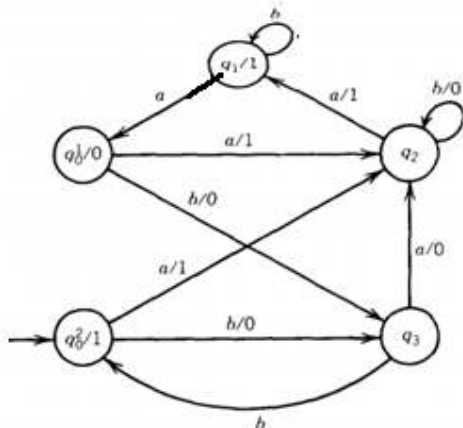


- We can begin the conversion process anywhere because the algorithm does not specify the order of replacing states; so let us first consider the state q_0 . Two edges come into this state, **one labeled $a/0$ and one labeled $b/1$** .
- Therefore, we need two copies of this state: one that prints a 0 (called q_0^1) and one that will print a 1 (called q_0^2).
- **Both of these states must be connected to q_2 through an edge labeled $a/1$ and to q_3 through an edge labeled $b/0$** . There is no loop at q_0 , so these two states are not connected to each other. The machine becomes:

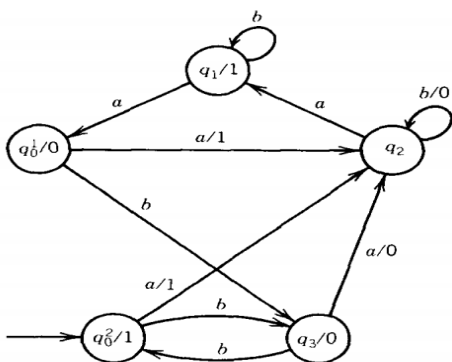


- We must select the start state for the new machine, so let us arbitrarily select q_0^2 .
- Notice that we now have two edges that cross. This sometimes happens, but aside from making a messier picture, there is no real problem in understanding which edge goes where.
- Notice that the edge from q_1 to q_0 , which used to be labeled $a/0$, is now only labeled a because the instruction to print the 0 is found in the state $q_0^1/0$.

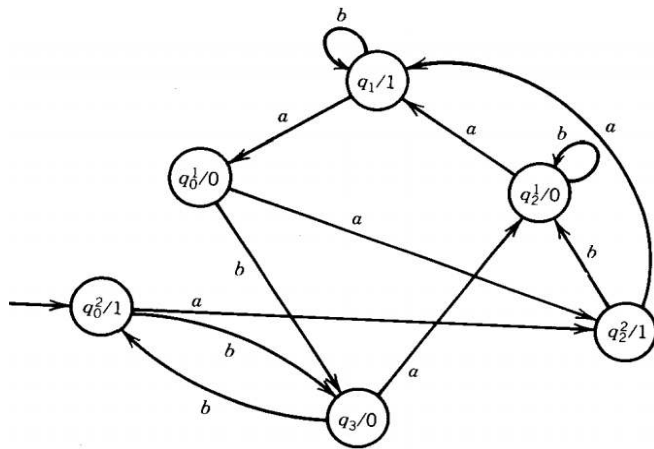
- The same is true for the edge from q_3 , which also loses its printing instruction.
- State q_1 has only two edges coming into it: one from q_2 labeled $a/1$ and a loop labeled $b/1$.
- So whenever we enter q_1 we are always printing a 1. We have no trouble here transferring the print instructions from the edges into the state. The machine now looks like this:



- What we have now is a partially converted machine or a hybrid. We could run an input string on this machine, and it would give us the same output as the original Me.
- The rules are that if an edge says print, then print; if a state says print, then print. If not, don't.
- Let us continue the conversion. State q_3 is easy to handle. Two edges come into it, both labeled $b/0$, so we change the state to $q_3/0$ and simplify the edge labels to b alone.

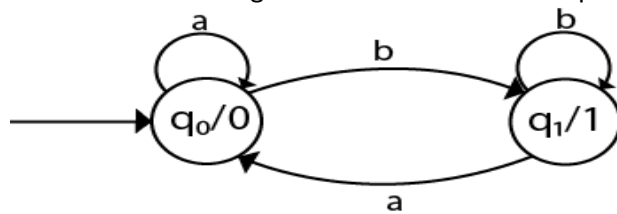


- The only job left is to convert state q_2 . It has some 0-printing edges entering it and some 1-printing edges (actually two of each, counting the loop).
- Therefore, we must split it into two copies, q_2^1 and q_2^2 . Let the first print a 0 and the second print a 1.
- **The two copies will be connected by a b edge going from q_2^2 to q_2^1 (to print a 0).** There will also be a b loop at q_2^1 . The final machine is this:

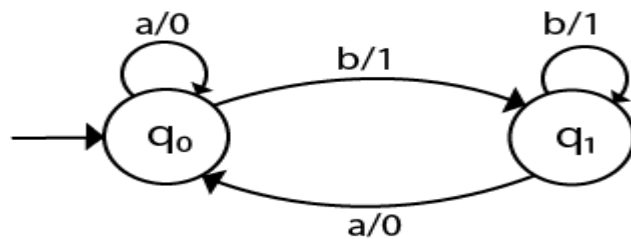


Example 2:

Convert the following Moore machine into its equivalent Mealy machine.



The equivalent Mealy machine will be,



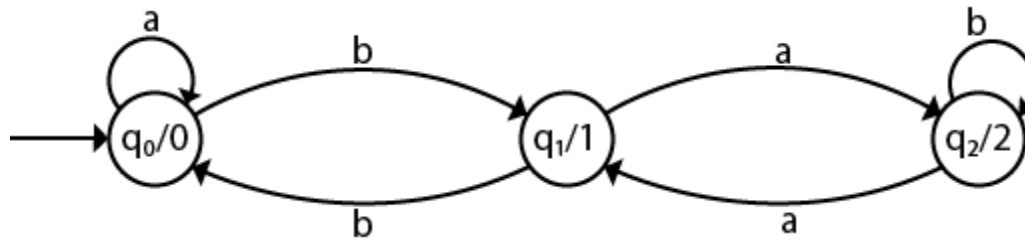
Example 3:

Convert the given Moore machine into its equivalent Mealy machine.

Q	a	b	Output(λ)
q0	q0	q1	0
q1	q2	q0	1
q2	q1	q2	2

Solution:

The transaction diagram for the given problem can be drawn as:



The equivalent Mealy machine can be obtained as follows:

